

ENV-LOGGER

Simple Overview

In simple language, this project

Connects to wifi → sets the time from NTP servers → adjusts the RTC according to it → collects data from sensors, 1 reading per minute → takes their rolling average over 5 minutes → displays them on the OLED display → logs data into SD card in CSV format and repeats it → then creates a new file after the day ends.

After setting the time once, the system is fully capable of working entirely offline!

Technical Overview

- **Standalone Node:** Acts as a self-contained, fault-tolerant environmental data logger with minimal dependency on network availability.
- **Timekeeping:** Uses NTP to fetch accurate time and syncs it to an RTC, ensuring reliable timestamps even during power or network failures.
- **Sensor Interface:** Collects data from multiple sensors—DHT (temp&humidity), BMP180 (pressure), and MQ135 (air quality)—via I2C and analog inputs.
- **Data Processing:** Implements a moving average filter (window size = 5) to reduce noise and stabilize readings.
- **Storage:** Logs data to an SD card over SPI in CSV format, with **daily file creation (YYYY-MM-DD.csv)** for efficient data management.
- **Output Channels:** Displays real-time data on OLED and streams via Serial for debugging.

The Motivation

1. Explore sensor fusion, figuring out how to make different modules (temperature, pressure, gas) play nicely together on a single microcontroller.
2. Master robust data logging, ensuring that data isn't just collected, but safely stored, formatted, and easily retrievable without data loss over long periods.

PINOUTS

DHT 11	ESP32
VCC	3V3
GND	GND
DATA	GPIO 04

BMP180	ESP32
VCC	3V3
GND	GND
SDA	GPIO 21
SCL	GPIO 22

RTC DS3231	ESP32
VCC	3V3
GND	GND
SDA	GPIO 21
SCL	GPIO 22

OLED SSD1306	ESP32
VCC	5V

GND	GND
SDA	GPIO 21
SCL	GPIO 22

MQ135	ESP32
VCC	5V
GND	GND
AOUT	GPIO 34

SD CARD MODULE	ESP32
VCC	3V3
GND	GND
MISO	GPIO 19
MOSI	GPIO 23
SCK	GPIO 18
CS	GPIO 15

How It Works & How to Use It

The Logging Process:

Once powered, the system tries to get a timestamp from NTP with the help of the internet through WIFI. Once the RTC is synced, the main loop starts.

Every 60 seconds, it wakes the sensors, takes a reading and pushes that value into an array. Once 5 readings are collected, it calculates the average, updates the OLED display and logs a single row to the day's CSV file on the SD card. When the RTC hits 00:00:00, it closes the current file and generates a new one for the next day.

Getting Started (Code Changes):

Before uploading the code to the board, you need to configure a few user-specific variables at the top of the sketch:

- **WiFi Credentials:** Enter your SSID and PASSWORD so the board can initially hit the NTP server.
- **Calibration Factors:** Adjust the MQ135 baseline R0 value or any specific temperature offsets depending on your exact sensor batch.
- **Timezone Offset:** Set your GMT offset (in seconds) for the NTP client to ensure the RTC matches your local time.

Accessing Data:

accessing the data is very simple:

1. Pop the MicroSD card out of the module.
2. Plug it into a card reader and connect it to your PC.
3. Open the .csv files directly in Excel, Google Sheets or Python for analysis!

Future Enhancements: A few features could take it to the next level:

- **Deep Sleep Integration:** Putting the microcontroller to sleep between the 1-minute read intervals to improve battery life significantly for off-grid solar deployments.
- **Wireless Data Offloading:** Adding a routine to periodically upload the day's CSV file to a local server (via MQTT or FTP) when WiFi is available, so the SD card never needs to be manually pulled.
- **Web Dashboard:** Building a lightweight local web server on the ESP32 to view live graphs from your phone.

- **Bye-bye!**

Thanks for checking out the project. Happy making, and may your code compile on the first try!

RESULTS AND ...

```
8 #include "SD.h" // SD card support
9 #include <SPI.h> // SPI communication (SD card)
10 #include <RTClib.h> // RTC library
11
12 #define DHTPIN 4 // DHT data pin
13 #define DHTTYPE DHT11 // Sensor type
14 #define SD_CS 13 // SD chip select
15 #define AVG_WINDOW 5 // Moving average window size
16 #define MQ135_PIN 34 // Analog pin for MQ135
17 #define RL_VALUE 30000.0 // Load resistance (Ohm)
18 #define RS_VALUE 244510.0 // Sensor baseline resistance
19 #define ROOM_BASELINE 24.00 // Reference baseline value
20
21 const char *ssid = " ";
22 const char *password = " ";
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24 DHT dht(DHTPIN, DHTTYPE); // DHT object
25 Adafruit_MQ135 mq135; // MQ object
26 RTC rtc; // RTC object
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